

Mobile Game Analytics: Player Behaviour, Retention & Monetization

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1. Executive Summary

Confidential Mobile Game Publisher needed a better way to analyze player engagement, retention, monetization, and geographic performance using data collected from multiple analytics platforms. This project developed a business intelligence solution that combined data from Firebase Analytics, GameAnalytics, and Google AdMob into a single reporting environment.

The project began by preparing the raw datasets in Python using the pandas and NumPy libraries. The data preparation process included validation, date standardization, feature engineering, and exporting four analysis-ready datasets covering engagement, player behaviour, monetization, and geography. These datasets were then imported into Power BI Desktop to create an interactive dashboard consisting of four report pages.

The completed dashboard answered the project's analytics questions by presenting key performance indicators and interactive visualizations that supported engagement analysis, retention monitoring, monetization evaluation, and geographic market comparisons. Functional testing confirmed that the prepared datasets loaded correctly, KPI calculations matched the source data, and all dashboard interactions performed as expected. All test scenarios were completed successfully, demonstrating that the solution satisfied the project's business and functional requirements.

Beyond confirming that the solution works, analysis of the completed dashboard surfaced several actionable business insights: player engagement is gradually eroding over the year, 85.5% of new players churn after their first day, revenue is almost entirely ad-dependent (99.6%), and revenue-per-player varies significantly by country. These findings are translated into specific, prioritized recommendations for the publisher.

The project also identified several opportunities for future improvement. These include replacing manual CSV exports with automated API connections, enabling scheduled dashboard refreshes through Power BI Service, introducing player-level analytics when more detailed data becomes available, and adding automated data quality validation to the Python workflow. These enhancements would improve the scalability, maintainability, and long-term usefulness of the solution.

2. Introduction

This report presents the project completed for the publisher. It will describe the development of a business intelligence solution that uses data collected from Firebase Analytics, GameAnalytics, and Google AdMob to analyze player engagement, retention, monetization, and geographic performance. The report will explain how the project progressed from understanding the business problem to designing, implementing, testing, and evaluating the final solution.

Section 3 will describe the business problem faced by the publisher and explain why improved reporting is important for supporting business decisions. Section 4 will present the analytics questions that guided the project, while Section 5 will define the project scope by outlining the included deliverables, project boundaries, assumptions, and limitations.

Section 6 will introduce the project's data sources, key data entities, and data flow. Section 7 will describe the data preparation process completed in Python, including data inspection, validation, cleaning, feature engineering, and the creation of the final analysis-ready datasets.

Section 8 will explain the proposed solution, its architecture, technology stack, dashboard design, and how it fits into the publisher's existing analytics environment. Section 9 will describe how the Python workflow and Power BI dashboard were implemented and how the completed solution was tested to verify that it met the project requirements.

Section 9 will also interpret the completed dashboard to surface key business insights about player engagement, retention, monetization, and geography, along with practical recommendations for the publisher. Finally, Section 10 will discuss potential improvements that could further enhance the solution through better performance, dashboard usability, data modelling, and future enhancements. The report will conclude with the references and appendices containing supporting material used throughout the project.

3. Business Problem Overview

The publisher collects large amounts of player engagement, gameplay, monetization, and geographic data from Firebase Analytics, GameAnalytics, and Google AdMob. Although these platforms provide valuable information, the data is stored across separate systems, making it difficult to obtain a complete view of player behaviour and overall business performance. This makes it challenging to monitor key performance indicators, compare market performance, identify player retention trends, and evaluate revenue generated through advertising and in-app purchases.

This challenge affects product managers and business stakeholders who rely on accurate reporting to evaluate player engagement, retention, monetization performance, and geographic trends when making business decisions. Without a centralized reporting solution, valuable insights remain scattered across multiple exported datasets, increasing the time and effort required to analyze performance.

To address this problem, the project developed a business intelligence solution that combines data from Firebase Analytics, GameAnalytics, and Google AdMob into a single reporting workflow. Python was used to prepare and validate the data, while Power BI was used to develop interactive dashboards that present player engagement, player behaviour, monetization, and geographic performance in a clear and consistent format.

The completed solution provides a centralized reporting environment that enables stakeholders to monitor important business metrics, evaluate player retention and revenue performance, compare results across different countries, and answer the analytics questions identified during the earlier stages of the project. By bringing together data from multiple sources into a single dashboard, the solution supports faster reporting and more informed business decision-making.

4. Analytics Questions

The following analytics questions guided the design and implementation of the project. They were developed to address the publisher's business objectives by examining player engagement, retention and churn, monetization performance, and geographic market performance. Each question is supported by one or more of the prepared datasets and is answered through the corresponding Power BI dashboard pages.

1. How engaged are players based on active users, sessions, and playtime?

This descriptive question evaluates overall player engagement using the engagement and player behaviour datasets. Daily, weekly, and monthly active users, together with session and playtime metrics, provide an overview of player activity and engagement trends over time. These metrics are presented on the Executive Overview and Player Behaviour dashboard pages to support performance monitoring.

2. Why are players leaving, and how do retention and churn patterns change across cohort days?

This diagnostic question examines player retention and churn using the player behaviour dataset. Retention and churn rates are analysed across install cohorts to identify where player drop-off is most significant during the early stages of the player lifecycle. The results are presented on the Player Behaviour dashboard page to support retention analysis.

3. Where does revenue come from, and how effectively does the game generate revenue through purchases and advertising?

This descriptive question evaluates the game's monetization performance using the monetization dataset. Revenue from advertising and in-app purchases is analysed alongside key performance indicators such as ARPU, ARPPU, eCPM, purchaser rate, and click-through rate

(CTR). These metrics are presented on the Monetization dashboard page to evaluate revenue performance and advertising effectiveness.

4. Which countries and market tiers contribute most to revenue and user value?

This descriptive question investigates geographic performance using the geography dataset. Country-level metrics, including active users, revenue, and Average Revenue per User (ARPU), are analysed together with the revenue-based market tier classification created during data preparation. These measures are presented on the Geography dashboard page to identify high-value markets and support business development decisions.

5. Scope Statement

This project developed a business intelligence solution to help the publisher analyze player engagement, retention, monetization, and geographic performance from multiple analytics platforms, covering data preparation, dashboard development, and testing. The sections below define what was included, excluded, and assumed.

5.1 In Scope

The project covers data from Firebase Analytics, GameAnalytics, and Google AdMob: nine raw CSV files consolidated into four analysis-ready datasets (engagement, player behaviour, monetization, geography), cleaned and prepared in Python, then built into a four-page interactive Power BI dashboard with KPIs and visualizations. Scope also includes testing dashboard functionality, data accuracy, and user interaction, plus recommendations for future improvement.

5.2 Out of Scope

Out of scope: predictive or machine learning analytics (the project is descriptive only); live connections to the three source platforms (the solution uses exported CSV files); Power BI Service deployment, mobile optimization, and role-based security (the dashboard is a desktop .pbix file); and any changes to the publisher's existing analytics platforms.

5.3 Constraints and Assumptions

The data is aggregated with no individual player identifiers, so analysis is limited to summarized metrics, and the four datasets sit at different levels of detail with no shared key, so each was analyzed separately. The solution assumes the exported CSVs accurately represent player activity and revenue, and was built within an academic 's time and resource limits using static exports rather than live connections. Analysis covers one year of historical data (2025); multiple years would support longer-term trend and seasonal analysis in future work.

6. Data Sources, Key Data Entities and Flows

This section describes the project's data sources, key data entities, and the flow of data from the original platforms to the final Power BI dashboard. Combining three analytics platforms, each covering a different aspect of player activity, monetization, and geography, gives a more complete view than any single platform alone.

6.1 Data Sources

The project uses data exported from Firebase Analytics (player engagement), GameAnalytics (gameplay behaviour), and Google AdMob (advertising revenue and geographic distribution). The exported CSV files are the input to the Python data preparation workflow before being imported into Power BI for reporting.

Data Source	Purpose	Key Fields	Limitations
Firebase Analytics	Provides player engagement and gameplay activity metrics.	Date, Active Users, Sessions, Screen Views, Average Session Duration	Exported as historical CSV files rather than live data.
GameAnalytics	Provides player behaviour and retention metrics based on cohort analysis.	Cohort Day, Retention Rate, Churn Rate, Average Playtime	Aggregated data does not include individual player identifiers.
Google AdMob	Provides monetization and advertising revenue information.	Week, Revenue, Ad Impressions, eCPM	Revenue is limited to advertising income only.
Geography Dataset	Provides player distribution and revenue by country.	Country, Active Users, Revenue, ARPU, Market Tier	Small player populations in some countries can affect ARPU values.

6.2 Key Data Entities

Four key data entities support the analysis: engagement (reporting dates, active users, sessions, average session duration), player behaviour (cohort-based retention rate, churn rate, average playtime), monetization (advertising revenue, ad impressions, eCPM), and geography (country, active users, revenue, ARPU, market tier). A complete description of all variables is provided in the Data_Dictionary.xlsx file (Appendix A).

6.3 Data Flow

Nine raw CSV files exported from the three platforms were manually reviewed and consolidated into four business-oriented datasets, which became the input to the Python data preparation workflow shown in Figure 6.1.

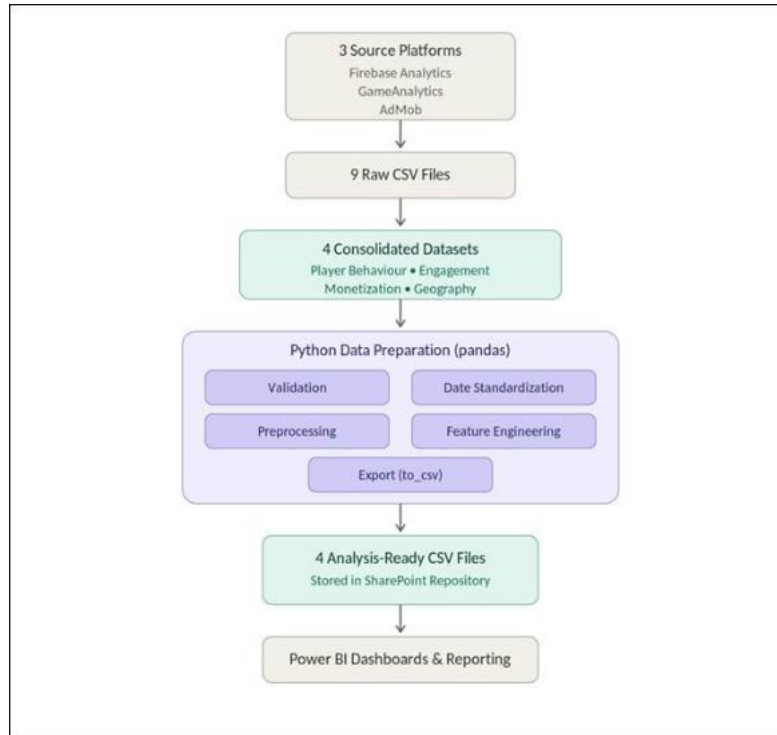


Figure 6.1. Data Flow of the Analytics Solution

The Python workflow (Jupyter Notebook) validates, cleans, standardizes, and prepares the four datasets, exporting them as analysis-ready CSV files stored in the repository. These are then imported into Power BI Desktop to build the interactive dashboards stakeholders use to monitor KPIs, identify trends, and support business decisions, without changing the original source systems.

7. Data Manipulation Process and Data Output

7.1 Data Inspection

Before any cleaning was performed, each of the four consolidated datasets—engagement, geography, monetization, and player_behaviour—was inspected to understand its structure and confirm that it had loaded correctly. The inspection reviewed each dataset's dimensions (shape), the first few records (head()), structural information and column data types (info()), and summary statistics for the numerical variables (describe()).

This initial inspection confirmed that all four datasets were imported successfully and provided an understanding of their structure before cleaning began. The datasets varied considerably in size, reflecting their different levels of granularity, including daily engagement metrics, country-level summaries, weekly monetization records, and cohort-day player behaviour data. Table 7.1 summarizes the characteristics of each dataset.

Table 7.1. Initial Dataset Characteristics

Dataset	Rows	Columns	Granularity
Engagement	365	11	Daily
Geography	92	12	Country
Monetization	53	35	Weekly
Player_behaviour	3,123	15	Cohort x Day

7.2 Data Cleaning and Transformation

All data cleaning and transformation was performed in Python using pandas. Since the nine source files were already consolidated into four business-oriented datasets, this stage focused on validating data quality and standardizing formats: duplicate checks, missing-value checks, data type validation, a review of text fields, and an assessment of irrelevant values.

Duplicate Records

Each dataset was checked for fully duplicated rows using `duplicated()`; no duplicates were found in any of the four datasets, so no rows were removed.

Missing Values

Missing values were assessed using `isnull()`; none were found, so no imputation, replacement, or row removal was required.

Data Type Validation

Three date fields, `date` (engagement), `week_start_date` (monetization), and `cohort_date` (player_behaviour), were imported as text and converted to datetime using pandas' `to_datetime()` function to support accurate time-based filtering, sorting, grouping, and visualization (Table 7.2).

Dataset	Column	Original Data Type	Updated Data Type
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Engagement	date	String (str)	Datetime (datetime64)
Monetization	week_start_date	String (str)	Datetime (datetime64)
Player_behaviour	cohort_date	String (str)	Datetime (datetime64)

Table 7.2 – Date type conversions (date, week_start_date, cohort_date)

Typographical Errors

Text fields were reviewed for spelling and formatting inconsistencies: country and country_code in the geography dataset matched consistently (e.g., United States/US), and acquisition_channel in the monetization dataset held a single value ("Direct"), so no corrections were required.

Irrelevant Values

Every field was reviewed and found to contribute to at least one business question, so no variables or records were removed.

Overall, the four datasets required only date-type standardization, confirming a clean, consistent foundation for feature engineering and export.

7.3 Data Quality Challenges

The main challenge was that three date fields were imported as text rather than datetime, resolved with pandas' to_datetime() as noted in Section 7.2; no formal outlier treatment was needed since the datasets consist of aggregated business metrics rather than raw transactional records. Because only minor standardization was required, effort focused primarily on feature engineering rather than extensive cleaning.

7.4 Final Data Outputs

A feature-engineering step then generated derived analytical fields from existing columns, without altering the underlying observations:

The engineered fields included:

- engagement: returning_pct (returning users as a percentage of total users), together with month and quarter fields derived from the date column to support time-based grouping and filtering.
- geography: arpu (Average Revenue per User, calculated as total revenue divided by active users) and market_tier, a revenue-based classification that groups countries into

Low, Medium, and High value tiers using the 25th and 75th percentiles of total revenue as thresholds.

- monetization: arpu (Average Revenue per User), together with month and quarter fields derived from week_start_date.
- player_behaviour: month and quarter fields derived from cohort_date to support cohort-based time analysis.

Each dataset was exported as a cleaned, analysis-ready CSV file using to_csv(index=False); exports were verified against expected row and column counts (see Appendix B), summarized in Table 7.4.

Dataset	Rows	Columns	Engineered Fields Added
Engagement	365	14	returning_pct, month, quarter
Geography	92	14	arpu, market_tier
Monetization	53	38	arpu, month, quarter
Player_behaviour	3,123	17	month, quarter

The four exported datasets were uploaded to the repository as the direct source for the Power BI implementation in Section 9.2.

7.5 Data Dictionary

A comprehensive data dictionary defines the variable name, data type, example value, and description for every dataset (Appendix A), covering:

The project includes separate data dictionaries for the following datasets:

Dataset	Description	Number of Variables
Player Behaviour	Cohort-based player retention, churn, and gameplay metrics.	15
Engagement	Daily engagement metrics, including active users, sessions, and playtime.	11
Monetization	Revenue, advertising, ARPU, ARPPU, LTV, and monetization performance metrics.	35
Geography	Country-level revenue, ARPU, and market-tier information.	12

8. New Solution Design and Fit into Existing IT Architecture

8.1 Proposed Solution

The proposed solution turns the publisher's exported player data into an interactive Power BI dashboard through the two-stage pipeline introduced in Section 6.3: Python-based preparation followed by Power BI reporting, using ten DAX measures across four report pages built from the four analysis-ready datasets.

The dashboard is organized into four modules aligned to the project's analytics questions: Executive Overview (high-level KPIs and trend lines), Player Behaviour (retention and churn by install cohort), Monetization (ad versus in-app revenue and yield metrics such as ARPU, ARPPU, and eCPM), and Geography (activity and revenue by country and market tier). Together, these replace the manual, platform-by-platform review the publisher previously relied on.

8.2 Solution Architecture

The solution architecture consists of six components arranged in a linear pipeline that moves from raw platform data to business decision-making, shown in Figure 8.1 and described below.

Data Sources. Firebase Analytics, GameAnalytics, and Google AdMob generate the player engagement, behavioural, and advertising data described in Section 6.1. Each platform is queried independently and is not connected directly to the rest of the pipeline.

CSV Files. Nine raw CSV files are exported and consolidated into the four business-oriented datasets described in Section 6.3. This export-based integration keeps the publisher's production analytics environment untouched.

Python/Pandas. The Jupyter Notebook workflow detailed in Section 9.1 validates, cleans, standardizes, and enriches the consolidated datasets, adding engineered fields such as ARPU, market tier, and returning-user percentage.

Prepared CSVs. The four analysis-ready CSV files, `engagement_clean`, `player_behaviour_clean`, `monetization_clean`, and `geography_clean`, are stored in the repository as the standardized interface between data preparation and reporting.

Power BI. The four datasets are imported into Power BI Desktop as fact tables connected through a shared Calendar dimension in a star schema, detailed further in Section 9.2, and turned into four report pages by ten DAX measures.

Business Users. the publisher stakeholders, including product managers and business decision-makers, use the completed dashboard to monitor KPIs and make data-informed decisions without needing to touch the underlying platforms or code.

8.3 Tools and Technology Stack

The solution was built using a small, complementary set of tools chosen for their fit with the project's aggregated, file-based data and the team's existing skills. Table 8.1 summarizes each tool, its role in the solution, and the reason it was selected.

Table 8.1. Tools and Technology Stack Summary

Tool	Role in Solution	Reason Selected
Python/Pandas	Data cleaning and transformation	Selected for efficient data cleaning, preprocessing, and transformation of raw datasets into analysis-ready CSV files.
CSV	Intermediate data storage	Selected as a lightweight and compatible file format for exchanging cleaned data between Python and Power BI.
Power BI	Dashboard and visualization	Selected for creating interactive dashboards, KPI visualizations, and business insights to support decision-making.
OneDrive	Collaboration and file storage	Selected to securely store project files, share datasets and reports, and enable collaboration among team members

8.4 Dashboard Design

The dashboard is organized into four report pages, shown below with their key visuals and how each supports the project's analytics questions.



Figure 8.2. Executive Overview dashboard page



Figure 8.3. Player Behaviour dashboard page

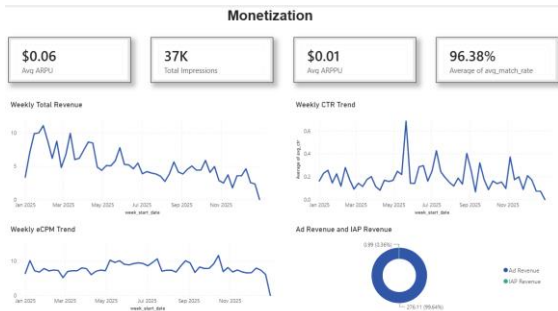


Figure 8.4. Monetization dashboard page

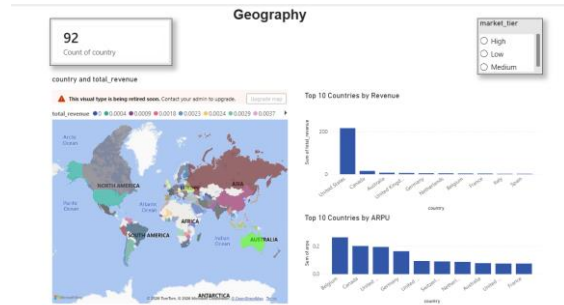


Figure 8.5. Geography dashboard page

Executive Overview displays six KPI cards (Avg ARPU, Day-7 Retention, Avg eCPM, DAU/MAU Ratio, Day-1 Retention, Total Revenue) with trend lines for active users, new-versus-returning players, and weekly revenue; the downward drift in these trends supports the engagement decline discussed in Section 9.5.1. Player Behaviour shows Day-1, Day-7, and Day-20 Retention cards alongside the retention curve, churn rate, and error rate by cohort day, where the steep drop between day 0 and day 1 is the churn cliff discussed in Section 9.5.2. Monetization tracks Avg ARPU, Total Impressions, Avg ARPPU, and match rate cards with weekly revenue, CTR, and eCPM trends plus an ad-versus-IAP donut confirming the ad-dependency finding in Section 9.5.3. Geography presents a Countries Represented card, a world revenue map, Top 10 Countries by Revenue and by ARPU bar charts, and a market-tier slicer; the map uses categorical rather than continuous colouring (Section 9.2), with the bar charts as the primary evidence for Section 9.5.4.

8.5 Fit with Existing IT Architecture

The analytics solution integrates with the publisher's existing environment rather than replacing any part of it. Firebase Analytics, GameAnalytics, and AdMob continue to operate independently and untouched; the integration point is the CSV export functionality each platform already provides, so the solution builds on exported copies rather than connecting to the live systems directly.

Once exported, the datasets are prepared using Python and pandas before being imported into Power BI for dashboard development, producing four analysis-ready datasets as the standardized input. This approach carries two practical benefits: there is no risk to the live analytics platforms or dependency on direct system integration, and the solution uses tools already familiar to the team, keeping it straightforward to maintain and repeat as new data is exported.

8.6 Expected Business Benefits

The completed solution delivers five concrete benefits to the publisher: it centralizes engagement, retention, monetization, and geographic reporting into one dashboard instead of three separate platform exports; it speeds up reporting by letting stakeholders open the dashboard rather than manually combining CSV exports; it surfaces the cohort-based retention curve and Day-1 churn cliff that were previously buried in raw GameAnalytics exports; it gives clear monetization visibility into advertising versus in-app revenue and yield metrics like eCPM

and ARPPU; and it supports geographic decision-making by ranking countries and market tiers on revenue and ARPU.

9. New Solution Implementation and Outcome Testing

9.1 Python Implementation

The data preparation stage was implemented in a Jupyter Notebook using Python, with pandas and NumPy supporting validation, preprocessing, and export. Python was chosen for its flexible, reproducible environment, and the notebook format lets each step be executed, verified, and repeated as new data becomes available.

Implementation Process

The implementation loads the four consolidated datasets into pandas DataFrames, engagement, geography, monetization, and player_behaviour, then inspects each to verify successful import and understand its structure, data types, and characteristics. Validation checks identify duplicate records, missing values, data type inconsistencies, typographical errors, and irrelevant data. The three date fields are standardized to datetime to support reliable time-based filtering, sorting, and visualization, and text fields such as country names and acquisition channels are reviewed for consistent formatting.

Once validated, a preprocessing stage generates additional analytical features, ARPU, Market Tier, Returning User Percentage, and Month/Quarter attributes, while preserving the original observations. These engineered fields support downstream reporting without modifying the underlying source data.

Tools and Techniques

Pandas was used throughout to load, inspect, validate, transform, and export the datasets, while NumPy supported the numerical calculations in feature engineering. Validation used duplicated() for duplicate records, isnull() for missing values, and pd.to_datetime() to standardize dates. The revenue-based Market Tier classification uses quantile-based binning, where the 25th and 75th percentile of total revenue split countries into Low, Medium, and High value groups.

ARPU and percentage calculations use NumPy's divide() with a where condition, returning a null value whenever a denominator is zero. No zero denominators occurred in the current data, but this protects the workflow against future datasets.

Implementation Outputs

The completed workflow exports four standardized, analysis-ready CSV files: engagement_clean.csv (365 × 14), geography_clean.csv (92 × 14), monetization_clean.csv (53 ×

38), and player_behaviour_clean.csv (3,123 × 17), each verified against expected row and column counts.

The exported datasets are stored in the repository as the standardized interface between Python preparation and the Power BI implementation in Section 9.2, ensuring all dashboard visualizations draw from the same reliable source.

9.2 Power BI Implementation

Data Import and Data Model:

The four datasets from the Python preparation stage (player_behaviour_clean, engagement_clean, monetization_clean, geography_clean) were imported into Power BI Desktop as fact tables. Because the tables sit at different grains (daily, weekly, cohort-day, country) with no usable common key, they were not related directly; instead, a star schema connects each fact table many-to-one to a single shared Calendar dimension, enabling consistent date-based filtering across every page. Date fields were set to the Date type on import, and the country field was set to the Country data category to enable the map visual.

DAX Measures:

Ten DAX measures were created to power the KPI cards and charts used across the four report pages, covering revenue, retention, engagement, and churn. This table summarizes each measure and its purpose:

Measure	DAX Formula	Purpose / Page
Total Revenue	SUM(monetization_clean[total_revenue_usd])	Total weekly ad + IAP revenue (Page 1 & 3)
Avg ARPU	AVERAGE(monetization_clean[arpu])	Avg revenue per active user/week (Page 1 & 3)
Avg ARPPU	AVERAGE(monetization_clean[arppu])	Avg revenue per paying user (Page 3)
Avg eCPM	AVERAGE(monetization_clean[avg_ecpm_usd])	Ad yield per 1,000 impressions (Page 1 & 3)
Ad Revenue %	AVERAGE(monetization_clean[ad_revenue_pct])	Share of revenue from ads vs IAP (Page 3)
DAU/MAU Ratio	AVERAGE(engagement_clean[dau_mau_ratio])	Player stickiness (Page 1)
Day-1 Retention	CALCULATE(AVERAGE(player_behaviour_clean[retention_rate]), player_behaviour_clean[cohort_day]=1)	% of cohort active Day 1 (Page 1 & 2)
Day-7 Retention	CALCULATE(AVERAGE(player_behaviour_clean[retention_rate]), player_behaviour_clean[cohort_day]=7)	% of cohort active Day 7 (Page 1 & 2)
Day-20 Retention	CALCULATE(AVERAGE(player_behaviour_clean[retention_rate]), player_behaviour_clean[cohort_day]=20)	% of cohort active Day 20 (Page 2)
Churn Rate (Day 1)	1 - [Day-1 Retention]	Headline Day-1 churn figure (85.5%) (Page 2)

Page Construction and Visuals:

The dashboard is built as four report pages, each addressing a specific business question:

- Executive Overview: Six KPI cards (Total Revenue, DAU/MAU Ratio, ARPU, Day-1 Retention, Day-7 Retention, Avg eCPM), a monthly active users trend, weekly revenue trend, and new-vs-returning users bar chart.
- Player Behaviour: Three KPI cards (Day-1, Day-7, Day-20 Retention), a retention curve across cohort days 0–20, a cumulative churn rate chart, and an error rate by cohort day chart.
- Monetization: Four KPI cards (Avg ARPU, Avg ARPPU, Total Impressions, Avg Match Rate), weekly revenue, eCPM, and CTR trends, and an ad-vs-IAP revenue split donut chart.
- Geography: A Countries Represented KPI card, a world map of player distribution, Top 10 Countries by Revenue and by ARPU bar charts, and a market_tier slicer that cross-filters the page.

Note: the Geography map uses categorical rather than continuous colouring, due to a Power BI visual-type limitation; the two bar charts remain the primary evidence for the geographic business question.

Interactions:

Each page is self-contained: slicer and filter selections on one page do not carry over to others. Hovering displays exact values in a tooltip, and clicking a bar, slice, or map region cross-filters the page down to that item, clicking again clears it. On Geography, the market_tier slicer filters the whole page to High-, Medium-, or Low-tier countries at once.

Formatting and Publishing:

The dashboard is distributed as a single Power BI file (.pbix) rather than published to the Power BI Service, since publishing requires a Pro license. The file opens directly in Power BI Desktop, which is free and requires no login, and already contains the cleaned datasets, measures, and visuals. To refresh with new data, a team member reruns the Python cleaning workflow, replaces the four CSV files, and selects Home > Refresh, no changes to measures, layouts, or visuals are required.

9.3 Testing Strategy and Test Scenarios

The completed Power BI solution was evaluated using three functional test scenarios designed to verify dashboard functionality, data accuracy, and fulfillment of the project's business requirements, summarized in Table 9.0.

Test Scenario	Criteria Checked
Test 1 – Dashboard Loading & Navigation	Dashboard loads successfully in Power BI Desktop; all four report pages are accessible; visuals display without errors.
Test 2 – KPI & Visualization Accuracy	KPI values match the prepared datasets; charts display the correct business metrics; revenue, retention, and geography visuals were spot-checked against source data.
Test 3 – Filter & Dashboard Interaction	Slicers and filters work correctly; cross-filtering updates related visuals as expected; users can isolate and analyze specific market tiers and countries.

Table 9.0. Testing Strategy and Test Scenarios

9.4 Overall Testing Evaluation

All three test scenarios were completed successfully. The prepared datasets imported into Power BI without errors, and all four dashboard pages loaded correctly with their intended KPI cards and visualizations (Test 1). KPI values and visualizations were verified against the prepared datasets and implemented calculations, with all results matching as expected (Test 2). Dashboard interactions, including the market-tier slicer and cross-filtering on the Geography page, performed correctly with no display or interaction issues (Test 3). No issues were identified, and the implemented solution satisfies the project's functional and business requirements.

Overall Status: ☒ Pass ☐ Fail ☐ Pending Further Review

9.5 Dashboard Insights and Business Recommendations

Since this project is based on descriptive and diagnostic analytics rather than a predictive model, this section looks at what the finished dashboard tells us about the publisher and turns those findings into practical recommendations, also cross-checked by querying the four datasets directly in Databricks to compare engagement, retention, monetization, and geography side by side. Each part below covers one dashboard area: what it shows, what it means for the business, and what the publisher should do about it.

9.5.1 Engagement: A Gradual Year-Long Decline in Stickiness

The Executive Overview page tracks how many players are active each day, week, and month throughout 2025. On average about 6.3% of monthly players open the game on any given day (the DAU/MAU ratio), but that yearly average hides a trend: the ratio was closer to 7.0% in Q1 and had dropped to about 5.9% by Q4, a gradual, sustained decline rather than a single event.

This looks like a slow build-up, more players drifting away than new players becoming regulars.

Recommendation: track DAU/MAU quarterly rather than relying on the yearly average, which hides the decline, and invest in win-back efforts (push notifications, seasonal events, new content) alongside new-player acquisition so any further drop is caught early.

9.5.2 Retention and Churn: The Day-1 Cliff

The Player Behaviour page shows how many players come back over their first 20 days, based on 347 install cohorts. 85.5% of new players don't return the day after installing; among those who stick around, retention settles into a lower 20–30% range.

This connects directly to the engagement trend above: cohorts that retain better also play more often, so the two issues are really one. Most players are lost before becoming regulars, which is likely what's driving the year-long engagement decline.

Recommendation: improving the first day should be the top priority, revisit onboarding, tutorial pacing, and first rewards, and test changes through structured A/B tests, using the retention

curve to compare results before and after each change rather than only to monitor performance.

9.5.3 Monetization: A Near-Entirely Ad-Supported Game

the publisher runs almost entirely on ads: they make up 99.6% of revenue, while in-app purchases barely register. Of 53 weeks, 42 had zero purchases, and in the weeks that did see one, fewer than 3.5% of players bought anything, usually spending very little.

Unlike the typical free-to-play pattern of a small spender group funding the game, purchases here are simply rare, so the publisher depends almost entirely on ads, with nothing to fall back on if ad rates or the ad market slow down. Even a small amount of in-app purchase revenue would give the business a cushion it doesn't currently have.

Recommendation: before rolling out new in-app offers, find out why almost no one buys, whether purchase options are hard to find, prices are right for each region, and purchase prompts are well-timed. In the meantime, watch eCPM and fill rate closely as early warning signs, since revenue relies so heavily on ads.

9.5.4 Geography: Engagement-Revenue Mismatches and Market Prioritization

The Geography page breaks revenue and users down by country across Low, Medium, and High tiers, and comparing countries turns up real mismatches: the Netherlands has 48 active users but about \$4.23 in revenue, while Belgium has just 13 active users but a similar \$3.43, noticeably more revenue per player. Since almost all revenue comes from ads rather than purchases, these gaps are best read as differences in ad performance by region rather than differences in spending behaviour.

Recommendation: treat country-level revenue-per-user gaps mainly as an ad-performance question; countries with plenty of active users but low revenue per user, like the Netherlands relative to Belgium, are worth reviewing for ad setup, fill rate, and eCPM.

9.5.5 Data Reliability Considerations

The data was already checked for missing values and duplicates in Section 7 and came back clean. A closer look for this section found a few smaller items worth watching: some unusually long play sessions (over 60 minutes) and a few countries with active players but zero revenue. These aren't necessarily problems, they could be small, not-yet-monetized markets, but are worth monitoring.

Recommendation: add a simple, ongoing check that flags these cases automatically each time the data refreshes, so anything worth investigating doesn't get missed.

9.5.6 Summary of Recommendations

Table 9.1 below summarizes these recommendations by dashboard area.

Dashboard Area	Key Insight	Recommendation	Priority
----------------	-------------	----------------	----------

Executive Overview (Engagement)	DAU/MAU dropped from ~7.0% (Q1) to 5.9% (Q4) — gradual, not flat	Track the ratio quarterly, not just annually; invest in win-back alongside acquisition	High
Retention & Churn	85.5% of new players don't return day 1; stayers drive engagement	Improve day-1 onboarding, tutorial, and pacing; test changes via A/B tests	High
Monetization	99.6% of revenue is ads; 42 of 53 weeks had zero purchases	Find out why purchases are rare before new offers; watch eCPM and fill rate closely	High
Geography	Revenue per player varies widely by country (e.g., Netherlands vs. Belgium) — an ad-performance gap, not spending preference	Review ad setup and eCPM in high-user, low-revenue countries	Medium
Data Reliability	A few small markets show zero revenue; some session lengths look inconsistent	Add an automated flag for these cases on each refresh	Low

Even without a predictive model, comparing what each dashboard page shows against the others surfaces clear, practical next steps. Section 10 covers a different kind of recommendation: improvements to the analytics solution itself, such as performance, usability, and automation.

10. Potential Solution Optimization

The implemented analytics solution successfully met the project requirements and passed all functional testing. Although the current implementation provides accurate reporting and interactive dashboards, several improvements could further enhance the solution's performance, usability, scalability, and long-term maintenance. The following recommendations focus on practical enhancements that build on the existing solution rather than changing its overall design.

10.1 Performance Improvements

The current solution performs well with the available datasets; however, a few improvements could make the reporting process more efficient as the volume of data grows.

The dashboard currently relies on four cleaned CSV files generated through the Python workflow. As larger datasets become available, loading times can be reduced by removing unused columns before exporting the final datasets and keeping only the fields required for reporting. This would reduce memory usage and improve dashboard responsiveness without affecting the analytical results.

The Python workflow could also include additional automated validation checks before exporting the cleaned datasets. Although the current datasets contained no duplicate records or missing values, automatic validation of row counts, data types, date ranges, and unexpected changes in key metrics would help identify potential issues earlier and improve the reliability of future data updates.

10.2 Dashboard Optimization

The Power BI dashboard was designed to provide a clear view of player engagement, retention, monetization, and geographic performance. Future versions could further improve the user experience by making navigation and data exploration more flexible.

Additional report-level filters, drill-through pages, and richer tooltips could help users investigate specific trends without increasing dashboard complexity. Conditional formatting could also be applied to KPI cards so that important changes in performance are highlighted automatically, making key trends easier to identify.

The Geography page currently uses a standard map visual to display player distribution by country. As Power BI continues to introduce new mapping features, upgrading to more advanced map visuals could improve the presentation of geographic information while preserving the existing dashboard structure.

10.3 Data Model Improvements

The solution currently uses four fact tables at different grains, daily, weekly, cohort-based, and country-level, connected only through a shared Calendar dimension rather than to each other directly. This star-schema approach ensures accurate calculations and avoids incorrect relationships between tables that don't share a usable common key.

If future datasets include common identifiers or more detailed transactional data, the model could be expanded to include shared dimension tables, such as a Date dimension, allowing additional cross-table analysis while maintaining data consistency. A more connected data model would also simplify future dashboard development as new reporting requirements are introduced.

Consistent field naming and standardized metadata across all datasets would further improve maintainability and make future enhancements easier to implement.

10.4 Future Optimization Opportunities

Several opportunities exist to extend the solution beyond its current implementation.

One improvement would be replacing the manual CSV export process with direct connections to Firebase Analytics, GameAnalytics, and Google AdMob through their available APIs. This would automate data collection, reduce manual effort, and ensure that reports always reflect the latest available information.

Publishing the dashboard through Power BI Service would also improve accessibility by allowing stakeholders to access reports through a web browser while supporting scheduled data refreshes. This would remove the need to manually distribute updated Power BI Desktop files whenever new data becomes available.

As the publisher continues to collect more detailed player information, the solution could be expanded to support player segmentation, lifetime value analysis, churn prediction, and revenue forecasting. These advanced analytics capabilities would provide deeper business insights while building on the reporting foundation developed during this project.

Finally, automated monitoring of data quality and refresh processes could be introduced to notify users of missing data, unexpected changes, or refresh failures. This would improve the long-term reliability of the analytics solution and reduce the amount of manual maintenance required.

References

Google LLC. (n.d.). Firebase Analytics documentation.

<https://firebase.google.com/docs/analytics>

GameAnalytics. (n.d.). GameAnalytics documentation. <https://gameanalytics.com/docs>

Google LLC. (n.d.). AdMob help center. <https://support.google.com/admob>

Microsoft. (2026). Power BI documentation. <https://docs.microsoft.com/en-us/power-bi>

pandas development team. (n.d.). pandas documentation. <https://pandas.pydata.org/docs>

11. Appendix

Appendix A – Data Dictionary

[Data Dictionary.xlsx](#)

Appendix B – Python Code or Notebook Excerpts

Loading and inspecting the four datasets:

```
import pandas as pd

import numpy as np

engagement = pd.read_csv("_dataset/engagement.csv")

geography = pd.read_csv("_dataset/geography.csv")

monetization = pd.read_csv("_dataset/monetization.csv")

player_behaviour = pd.read_csv("_dataset/player_behaviour.csv")

print("Engagement:", engagement.shape)

print("Geography:", geography.shape)

print("Monetization:", monetization.shape)
```

```
print("Player Behaviour:", player_behaviour.shape)
```

```
display(engagement.head())
```

```
display(geography.head())
```

```
display(monetization.head())
```

```
display(player_behaviour.head())
```

```
engagement.info()
```

```
geography.info()
```

```
monetization.info()
```

```
player_behaviour.info()
```

Duplicate and missing value validation:

```
print("Engagement:", engagement.duplicated().sum())
```

```
print("Geography:", geography.duplicated().sum())
```

```
print("Monetization:", monetization.duplicated().sum())
```

```
print("Player Behaviour:", player_behaviour.duplicated().sum())
```

```
print(engagement.isnull().sum())
```

```
print(geography.isnull().sum())
```

```
print(monetization.isnull().sum())
```

```
print(player_behaviour.isnull().sum())
```

Date field standardization:

```
engagement["date"] = pd.to_datetime(engagement["date"])
```

```
monetization["week_start_date"] = pd.to_datetime(monetization["week_start_date"])
```

```
player_behaviour["cohort_date"] = pd.to_datetime(player_behaviour["cohort_date"])
```

Typographical / Consistency Checks

```
print(geography["country"].unique())
```

```
print(geography["country_code"].unique())
```

```
print(monetization["acquisition_channel"].unique())
```

Revenue-based Market Tier:

```
low = geography["total_revenue"].quantile(0.25)

high = geography["total_revenue"].quantile(0.75)

geography["market_tier"] = pd.cut(

    geography["total_revenue"],

    bins=[-np.inf, low, high, np.inf],

    labels=["Low", "Medium", "High"]

)

monetization["arpu"] = np.divide(

    monetization["total_revenue_usd"],

    monetization["active_users"],

    out=np.full(len(monetization), np.nan, dtype=float),

    where=monetization["active_users"].ne(0)

).round(4)

# Month and quarter

monetization["month"] = monetization["week_start_date"].dt.month

monetization["quarter"] = monetization["week_start_date"].dt.quarter

# Engagement

# Returning user percentage

engagement["returning_pct"] = np.divide(

    engagement["returning_users"] * 100,

    engagement["total_users"],

    out=np.full(len(engagement), np.nan, dtype=float),

    where=engagement["total_users"].ne(0)

).round(2)

# Month and quarter

engagement["month"] = engagement["date"].dt.month
```



```

engagement["quarter"] = engagement["date"].dt.quarter

# Player Behaviour

# Month and quarter from cohort (install) date

player_behaviour["month"] = player_behaviour["cohort_date"].dt.month

player_behaviour["quarter"] = player_behaviour["cohort_date"].dt.quarter


# Verify preprocessing

print("Calculated fields added successfully.\n")

print("Dataset shapes:")

print(f"Engagement:    {engagement.shape}")

print(f"Geography:      {geography.shape}")

print(f"Monetization:    {monetization.shape}")

print(f"Player Behaviour: {player_behaviour.shape}")

Exporting and verifying the analysis-ready datasets:

engagement.to_csv("engagement_clean.csv", index=False)
player_behaviour.to_csv("player_behaviour_clean.csv", index=False)
monetization.to_csv("monetization_clean.csv", index=False)
geography.to_csv("geography_clean.csv", index=False)

import os

exported_files = {

    "engagement_clean.csv": engagement,

    "player_behaviour_clean.csv": player_behaviour,

    "monetization_clean.csv": monetization,

    "geography_clean.csv": geography

}

for filename, df in exported_files.items():

    print(

        filename,

        "| exists:", os.path.exists(filename),

```

```
"| rows:", df.shape[0],

"| columns:", df.shape[1]

)
```

Supporting screenshots for the data inspection and validation steps described in Section 7:

```
[5]: print("Shape")
      print(engagement.shape)

      print("\nFirst 5 Records")
      display(engagement.head())

      print("\nDataset Information")
      engagement.info()

Shape
(365, 11)

First 5 Records
  date    day  monthly_active_users  weekly_active_users  daily_active_users  dau_mau_ratio  dau_wau_ratio  wau_mau_ratio  new_users
0 2025-01-01    0                227                 76                 16         0.070485         0.210526         0.334802             1
1 2025-01-02    1                227                 67                 8          0.035242         0.119403         0.295154             0
2 2025-01-03    2                224                 67                 11         0.049107         0.164179         0.299107             0
3 2025-01-04    3                223                 62                 9          0.040359         0.145161         0.278027             1
4 2025-01-05    4                225                 71                 19         0.084444         0.267606         0.315556             0

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 365 entries, 0 to 364
Data columns (total 11 columns):
#   Column              Non-Null Count  Dtype
---  -
0   date                 365 non-null   str
1   day                  365 non-null   int64
2   monthly_active_users 365 non-null   int64
3   weekly_active_users  365 non-null   int64
4   daily_active_users   365 non-null   int64
5   dau_mau_ratio        365 non-null   float64
6   dau_wau_ratio        365 non-null   float64
7   wau_mau_ratio        365 non-null   float64
8   new_users            365 non-null   int64
9   returning_users      365 non-null   int64
10  total_users          365 non-null   int64
dtypes: float64(3), int64(7), str(1)
memory usage: 35.1 KB
```

Figure B.1. Initial inspection of the engagement dataset (shape, head(), info()).

```
print("Engagement:", engagement.duplicated().sum())
print("Geography:", geography.duplicated().sum())
print("Monetization:", monetization.duplicated().sum())
print("Player Behaviour:", player_behaviour.duplicated().sum())

Engagement: 0
Geography: 0
Monetization: 0
Player Behaviour: 0
```

Figure B.2. Duplicate validation (duplicated() output).

```

print("Engagement:", engagement.isnull().sum().sum())
print("Geography:", geography.isnull().sum().sum())
print("Monetization:", monetization.isnull().sum().sum())
print("Player Behaviour:", player_behaviour.isnull().sum().sum())

Engagement: 0
Geography: 0
Monetization: 0
Player Behaviour: 0

```

Figure B.3. Missing value validation (isnull() output).

```

# export clean datasets

engagement.to_csv("engagement_clean.csv", index=False)
player_behaviour.to_csv("player_behaviour_clean.csv", index=False)
monetization.to_csv("monetization_clean.csv", index=False)
geography.to_csv("geography_clean.csv", index=False)

import os

exported_files = {
    "engagement_clean.csv": engagement,
    "player_behaviour_clean.csv": player_behaviour,
    "monetization_clean.csv": monetization,
    "geography_clean.csv": geography
}

for filename, df in exported_files.items():
    print(
        filename,
        "| exists:", os.path.exists(filename),
        "| rows:", df.shape[0],
        "| columns:", df.shape[1]
    )

engagement_clean.csv | exists: True | rows: 365 | columns: 14
player_behaviour_clean.csv | exists: True | rows: 3123 | columns: 17
monetization_clean.csv | exists: True | rows: 53 | columns: 38
geography_clean.csv | exists: True | rows: 92 | columns: 14

```

Figure B.4. Export and verification of the four analysis-ready datasets (to_csv()).